

❏ 1. Configuring the Apache Web Server

❏ What is Apache?

- Apache HTTP Server (**httpd**) is a widely-used open-source web server.
 - It serves static and dynamic web content over **HTTP/HTTPS**.
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❏ Installing Apache:

Debian/Ubuntu:

```
bash
```

Copy code

```
sudo apt install apache2
```

RHEL/CentOS:

```
bash
```

Copy code

```
sudo yum install httpd
```

⌘ Basic Directory Structure:

Path	Description
/etc/httpd/ or /etc/apache2/	Configuration files
/var/www/html/	Default document root

/etc/httpd/conf/httpd.conf	Main config file (CentOS)
/etc/apache2/apache2.conf	Main config file (Ubuntu)
/etc/apache2/sites-available/	Virtual host configs
/var/log/httpd/ or /var/log/apache2/	Logs

☒ Managing Apache:

bash

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```
sudo systemctl start apache2    # Ubuntu sudo systemctl restart httpd    # CentOS sudo systemctl enable apache2    # Enable on boot
```

☒ 2. Apache Security & Virtual Hosting

☒ Apache Security

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Disable Directory Listing:

```
apache
```

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Options -Indexes

-

Restrict access with .htaccess:

```
apache
```

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```
AuthType Basic AuthName "Restricted" AuthUserFile /etc/apache2/.htpasswd Require valid-user
```

-

Create .htpasswd:

bash

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```
htpasswd -c /etc/apache2/.htpasswd username
```

-

Disable unnecessary modules:

bash

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```
sudo a2dismod autoindex
```

-

Enable Firewall Rules:

```
bash
```

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```
sudo ufw allow 'Apache Full'
```

-

Run as non-root (default apache user)

📄 Virtual Hosting

Virtual Hosting allows multiple websites to be hosted on a single server.

📄 Name-Based Virtual Host Example:

apache

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```
<VirtualHost *:80> ServerAdmin admin@example.com ServerName example.com DocumentRoot /var/www/example.com ErrorLog  
${APACHE_LOG_DIR}/example.com_error.log </VirtualHost>
```

1.

Create Site Directory:

bash

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```
sudo mkdir -p /var/www/example.com
```

2.

Enable Site:

```
bash
```

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```
sudo a2ensite example.com.conf sudo systemctl reload apache2
```

3.

Disable Site:

```
bash
```


Copy code

```
sudo a2dissite 000-default.conf
```

❏ 3. Configuring the Squid Web Proxy Cache

❏ What is Squid?

- Squid is a **caching proxy server** for HTTP, HTTPS, FTP.
- Improves speed, security, and controls internet usage.

❏ Install Squid:

Debian/Ubuntu:

```
bash
```

Copy code

```
sudo apt install squid
```

RHEL/CentOS:

```
bash
```

Copy code

```
sudo yum install squid
```

☒ Main Config File:

- /etc/squid/squid.conf

☒ Basic Configuration Steps:

1.
Edit squid.conf:

conf

Copy code

```
http_port 3128 acl localnet src 192.168.1.0/24 http_access allow localnet
```

2.
Restart Squid:

```
bash
```

Copy code

```
sudo systemctl restart squid
```

3.

Enable Squid:

```
bash
```

Copy code

```
sudo systemctl enable squid
```

❏ Caching and Logs:

File	Description
/var/spool/squid	Cache directory
/var/log/squid/access.log	Access log
/var/log/squid/cache.log	Errors and events

❏ Squid ACLs (Access Control Lists):

conf

Copy code

```
acl office src 192.168.1.0/24 http_access allow office http_access deny all
```

🔒 Block Sites:

conf

Copy code

```
acl blocked_sites dstdomain .facebook.com http_access deny blocked_sites
```